# Evaluating Expressions with Radicals

*Lesson Plan*

## 1. Lesson Information

| | | | |
|---|---|---|---|
| Course | Grade 7 Accelerated (1205050) | Unit | 4 — Solving Problems with Rational Numbers |
| Lesson | Lesson 5 — Evaluating Expressions with Radicals | Date | _____ |
| Length | 53 minutes (one period) | Benchmark(s) | MA.8.NSO.1.7 |

## 2. Standards

**MA.8.NSO.1.7** Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

**Must.** Evaluate expressions containing square roots, cube roots and integer exponents, simplifying radicals by factoring within the benchmark's lists.

**Must not.** Perfect squares up to 225 and perfect cubes from −125 to 125, or fractions whose parts are each in those lists. Estimating a root that is not perfect was Unit 2. No fractional exponents.

## 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR | What it asks | The evidence in this lesson |
|---|---|---|
| MA.K12.MTR.5.1 | Use patterns and structure to help understand and connect mathematical concepts. | Notes I — the radical bar is recognised as a grouping symbol by comparing √(9 + 16) with √9 + √16, so the rule is derived from the difference rather than announced. |
| MA.K12.MTR.6.1 | Assess the reasonableness of solutions. | Boards 3 and 6 — students say whether a negative answer is possible at all before they evaluate, which is the whole square-root-versus-cube-root distinction. |
| MA.K12.MTR.4.1 | Engage in discussions that reflect on the mathematical thinking of self and others. | Board 5 — when the room splits between 13 and 17 on √(25 + 144), the sixty-second re-vote before the reveal. |

## 4. Learning Target and Essential Question

**Learning target.** I can evaluate an expression that contains square roots, cube roots and integer exponents.

**Essential question.** What does the radical bar do to the order in which an expression is evaluated?

**Building on.** Unit 2 (2.01–2.06) — perfect squares, perfect cubes and the principal square root; 3.04 — negative exponents; Grade 6 — the order of operations.

**Working toward.** 4.06 — the whole order of operations at once; 4.07–08 — real-world multi-step problems; MA.912.NSO.1.4 in Algebra 1.

## 5. Sequence

*read from the deck; never typed*

| Segment | Min | What the teacher does | What students do |
|---|---|---|---|
| Settle in; post the learning target | 1 | Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet. | Copy the target. |
| Warm-Up | 6 | Silent, then reveal. One sentence of reteach per question at most. Then the new words, read once. | Four retrieval questions, alone, no notes. Then copy the new words. |
| Notes | 12 | Teach from the front; say the off-slide sentence out loud. | Copy the notes; answer the checks aloud. |
| Example 1 | 3 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Example 2 | 3 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Whiteboards | 11 | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three. |
| Independent Set | 6 | Silent. Circulate and mark what you see; do not teach. | Six questions, written, alone. |
| IXL | 5 | Assign the day's IXL skills. | Start the skills; this skill continues tomorrow: one assignment, due at the start of the class after 4.06. |
| **Total** | **53** | | |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the off-slide sentence is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question | DOK |
|---|---|
| Why is $\sqrt{(9 + 16)}$ not the same as $\sqrt{9} + \sqrt{16}$, when $(9 + 16) \div 5$ is the same as $9 \div 5 + 16 \div 5$? | DOK 3 |
| A cube root of a negative number is ordinary and a square root of one is not. What is it about the exponent 2 that makes the difference? | DOK 3 |
| $\sqrt{49}$ is 7, but the equation $x^2 = 49$ has two solutions. Are those two statements in conflict? | DOK 3 |

## 8. Checks for Understanding

*each board, and what to do about it*

| # | What the board asks | Answer | If the board is wrong |
|---|---|---|---|
| 1 | $\sqrt{121}$ | ***11*** | 60.5 — the radicand was halved [Unit 2, 2.02: a root is not a half]; ±11 — the radical sign names the principal root only [8.NSO.1.7] |
| 2 | A square patio has an area of 169 square feet. Fencing for it costs 12 dollars a foot. How long is one side of the patio? | ***13 feet*** | anything built from 12 dollars — the unneeded figure was used [the question asks for a length, not a cost]; 52 — the perimeter, not a side; 84.5 — the area was halved |
| 3 | $\sqrt[3]{-125}$ | ***−5*** | 5 — the sign was dropped [8.NSO.1.7: an odd root keeps the sign of its radicand]; 'not a real number' — the square-root rule applied to a cube root [B1G-M 8.NSO.1.7 misconception: confusing square roots with cube roots] |
| 4 | $\sqrt{2^3 + 17}$ | ***5*** | 25 — the arithmetic under the bar was done and the root was not [8.NSO.1.7]; anything with two separate roots — the bar groups, so it is one radicand [8.NSO.1.7] |
| 5 | √(25 + 144). Which is the value? | ***D — 13*** | (A) the root of each piece was taken and the results added, 5 + 12 [8.NSO.1.7: the bar groups, so the radicand is 169]; (B) the sum was found and the root was never taken [8.NSO.1.7]; (C) the radicand was halved instead of rooted [Unit 2, 2.02: a root is not a half] |
| 6 | The cube root of −1/64. Which is the value? | ***A — −1/4*** | (B) the square-root rule for negatives was applied to a cube root [B1G-M 8.NSO.1.7 misconception]; (C) the square root of 1/64 was taken and the sign kept [B1G-M 8.NSO.1.7 misconception: confusing square roots with cube roots]; (D) the sign was dropped [8.NSO.1.7: an odd root keeps the sign of its radicand] |
| 7 | $\sqrt{\frac{4}{9}} + \sqrt[3]{\frac{8}{27}}$ | ***4/3*** | 12/36 — the roots were never taken, then 4/9 + 8/27 was added top-and-bottom [two errors: the roots, then Grade 5 fraction addition]; 2/3 — only one of the two radicals was evaluated |
| 8 | $\sqrt{5^3 - 25} \cdot 10^{-3}$ | ***1/100*** | 10,000 — $10^{-3}$ read as $10^3$ [8.NSO.1.3: a negative exponent is a reciprocal]; −1/100 — the exponent's sign made the value's sign [B1G-M 8.NSO.1.3 task: −b vs $b^{-1}$] |
| 9 | Simon says that the cube root of −27 is not a real number, because you cannot take the root of a negative. Explain why he is wrong, and give the value. | ***−3*** | '−3' alone — the value without the reason [the item asks why he is wrong]; '3' — the sign dropped as well as the rule misread |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

**ESE / IEP.** Give the perfect-square and perfect-cube lists from Unit 2's reference sheet and let them stay out; a root is a lookup today, not a recall. On every board with arithmetic under the bar, the student draws a box around the radicand before writing anything, so the grouping is a physical act.

**ELL.** Radicand and radical are the two new nouns, and they are easy to confuse because they sound alike. Write √25 once with RADICAL circling the whole symbol and RADICAND labelling only the 25, and point at the labels on every board.

**Enrichment.** Ask for a pair of numbers for which √(a + b) really does equal √a + √b, and what has to be true of them. Then ask why the same question for a product has a different answer.

## 10. Closure, Vocabulary, Materials, Practice

**Closure.** Board 9 is written work and it is the exit evidence: the student has to say which root Simon's rule belongs to, not only that he is wrong, and give −3. Read the boards, not the papers.

**Vocabulary.** **radical**; **radicand**; **principal square root**

**Materials.** Whiteboards and markers, and Unit 2's reference sheet if students still want the perfect-square list. Calculators are allowed on everything; a calculator will refuse the cube root of a negative on some models, which is worth showing once and explaining rather than trusting.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, this skill continues tomorrow: one assignment, due at the start of the class after 4.06: Evaluate radical expressions — D8J.